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The ballad of LLM agents: philosophical reasoning for chemistry

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E-mail: hharb@anl.gov and assary@anl.gov**Keywords:** large language models, agentic ai, scientific reasoning, philosophical promptingSupplementary material for this article is available [online](#)

Abstract

Large language models (LLMs) show remarkable potential for scientific reasoning but often produce unreliable or scientifically unactionable outputs when faced with multi-step logic, domain grounding, and interpretability challenges, especially in complex fields like chemistry and materials science. Here, we introduce a framework of philosophical reasoning agents, inspired by canonical thinkers such as Socrates, Descartes, Kant, and Hume, to guide LLM behavior via structured prompt engineering. These agents embody distinct reasoning paradigms (dialectical inquiry, deductive logic, rule-based judgment, and empirical validation) and are evaluated across multiple chemistry subdomains, physical, analytical, general, inorganic, and organic chemistry, using the ChemBench benchmark. Our agentic prompting approach yields substantial accuracy gains on open-ended numerical chemistry questions, with gains of +11.5 percentage points for GPT-4o with Hume, +4.5 percentage points for GPT-5 with Kant, and +21.8 percentage points for GPT-5.1 with Socrates at the strict 1% error threshold, relative to the corresponding base models. Beyond accuracy, we observe benchmark-level model-agent performance patterns, suggesting that different prompting styles interact differently with each base model. These findings demonstrate that embedding philosophy-of-science principles into multi-agent frameworks can improve and produce interpretable, adaptive, and domain-aligned scientific LLMs.

1. Introduction

Large language models (LLMs) are increasingly deployed as general-purpose reasoning engines across scientific domains, including chemistry and materials science [1–8]. Recent advances have demonstrated impressive performance on tasks ranging from reaction prediction and property estimation to literature analysis and hypothesis generation [1, 9–11]. Despite these gains, LLMs continue to exhibit systematic failures in scientific reasoning, particularly on tasks that require multi-step logic, numerical accuracy, or explicit grounding in physical and chemical principles [4, 5, 8, 12–17]. These limitations are especially evident when models are evaluated beyond binary correctness, where near-miss errors and unstable reasoning trajectories can undermine scientific reliability [1, 4, 11, 18].

To address these challenges, a growing body of work has explored *agentic* and *prompt-based* strategies aimed at improving reasoning behavior without modifying model weights [9, 17, 19–22]. Techniques such as chain-of-thought (CoT) prompting [23, 24], self-consistency [25], tool use [1, 26], and multi-agent orchestration have shown that structured guidance can substantially improve model outputs [27–29]. Recent work has also explored more specialized agentic reasoning and chemistry-agent frameworks, including multi-agent inference systems based on abductive, deductive, and inductive reasoning, cooperative agents for explainable quantum-chemistry property prediction, and physically grounded benchmarks for crystallographic reasoning [30–32]. However, many of these approaches rely on task-specific

heuristics, extensive orchestration, or opaque prompt engineering practices, making it difficult to reason about *why* a given strategy works or how it should be adapted across tasks and model architectures [1, 2, 9, 28]. Such behavior reflects the internal paradox of LLM development, between directly completing user’s prompts as quick as possible and exploring different layers of various domain-specific expert-like reasoning within a limited given context [4, 5, 16].

An alternative perspective is to view LLM reasoning through the lens of classical philosophy [21, 33–37]. Philosophical traditions have long formalized distinct modes of inquiry, including deductive reasoning, empirical induction, dialectical synthesis, and rule-based judgment, which explore the reasoning framework at a more abstract level [38–40]. Several studies have suggested that embedding such philosophical principles into system-level instructions can influence how LLMs reason, offering a principled and interpretable approach to agent design [36, 41–44].

Recent studies, including our own, have shown that structured Socratic-style guidance can substantially improve LLM performance on complex reasoning tasks, particularly in scientific domains that require explicit logic and interpretability [21, 36, 45]. In our prior work, we formalized Socratic prompting for chemistry by operationalizing ten core principles derived from Chang’s framework [36], such as definition, induction, elenchus (critical questioning through refutation), hypothesis elimination, and analogy, and demonstrated that these principles improve clarity, reduce ambiguity, and support hypothesis refinement [21]. We also extended this foundation by introducing a Mixed Socratic Prompting approach, which integrates multiple principles within a single reasoning workflow, moving beyond linear CoT toward structured, multi-perspective analysis aligned with but distinct from standard CoT prompting [21]. Building on these ideas, we proposed a Socratic method Agent that autonomously selects and applies Socratic strategies as ‘reasoning tools’, generates its own follow-up queries, and supports iterative, hypothesis-driven reasoning, a paradigm we refer to as chain-of-reasoning prompting, which is abstraction of reasoning patterns instead of traditional direct pattern fulfillment [45]. Our prior work demonstrated that Socratic-guided agents improve reasoning depth, hypothesis generation, and scientific framing across areas relevant to chemistry and materials science, including electrochemistry, physical chemistry, and materials degradation. The results of our previous study motivate a broader investigation into philosophy-guided reasoning as a general mechanism for stabilizing and enhancing scientific inference with LLMs [45]. The present study extends that earlier Socratic framework by asking whether other philosophy-inspired reasoning motifs can be operationalized as fixed system prompts and benchmarked under the same ChemBench numerical-answer setting. This allows us to move from a single Socratic reasoning framework to a comparative evaluation of multiple philosophy-inspired reasoning templates under controlled conditions.

In this work, we introduce a philosophy-of-science-inspired agentic framework for guiding LLMs in chemistry reasoning tasks. Drawing on the reasoning paradigms of canonical thinkers, like Socrates (dialogical inquiry), Descartes (deductive logic), Kant (rule-based judgment), and Hume (empirical validation), we construct seven philosophy agents that emulate these distinct epistemic styles. Our central hypothesis is that embedding such reasoning archetypes can enhance both performance and interpretability of LLMs in domain-specific scientific tasks. We conduct a controlled evaluation of these agents across three leading LLMs (GPT-4o, GPT-5, and GPT-5.1) [46, 47] on a subset of the ChemBench [4, 48] benchmark dataset, spanning multiple subfields of chemistry. We use the open-ended numerical-answer subset of ChemBench. This subset contains 243 questions annotated across five chemistry subdomains: physical, analytical, general, inorganic, and organic chemistry. The same questions are also annotated by task type, including reasoning, calculation, and knowledge. Results (*vide infra*) show that philosophical prompting yields significant improvements over zero-shot baselines, with performance gains of +18.4% for GPT-4o with Hume, +14.1% for GPT-5 with Kant, and +11.7% for GPT-5.1 with Descartes. Furthermore, model-agent pairings reveal consistent benchmark-level performance patterns, suggesting that different prompting styles interact differently with each base model. We interpret these patterns behaviorally and do not claim direct access to the models’ internal mechanisms. While this work focuses on benchmarking performance, it also lays the groundwork for developing future agent-selection heuristics based on task demands and model characteristics.

2. Methods

2.1. Construction of Philosophy Agents (PA)

Each Philosophy Agent (PA) was implemented as a fixed system prompt encoding a distinct philosophical reasoning framework. The agents were designed to shape *how* the base model reasons, rather than *what* information it accesses [45]. We did not use fine-tuning, retrieval augmentation, external tools, or

Table 1. Philosophical LLM reasoning tools explored in this work.

Philosopher	School/thought	Principles
Socrates	Classical Greek philosophy	<i>Socratic questioning; definition refinement; assumption testing; hypothesis elimination; step-by-step inquiry; self-critique; foundation of critical thinking</i>
Aristotle	Natural philosophy	<i>Categorization; first principles; causal reasoning; formal definitions; teleological explanation</i>
Descartes	Cartesian rationalism	<i>Decomposition; axiomatic deduction, stepwise reasoning; reduction to simples; methodical doubt; logical progression</i>
Hegel	German idealism	<i>Dialectical reasoning; thesis–antithesis–synthesis; contradiction resolution; conceptual development</i>
Hume	Empiricism	<i>Empirical grounding; example-driven reasoning; pattern recognition; inductive inference; experience-based justification</i>
Plato	Platonic idealism	<i>Abstract reformulation; generalized ideal forms/symmetries; conceptual ascent; value-oriented reasoning; follow-up questioning</i>
Kant	Transcendental idealism	<i>Premises-prioritized reasoning; constraints and conditions; structured inference; epistemic limits; synthetic organization</i>

memory mechanisms. The only difference between agents is the system-level instruction provided to the model. The philosophical strategies were grounded in well-established schools of thought, as listed in table 1: Socratic prompting emphasizes iterative questioning, clarification of assumptions, and explicit justification of intermediate steps [21, 38, 39, 45]. Aristotelian prompting focuses on causal structure, categorization, and explanation through first principles. Cartesian prompting enforces problem decomposition, axiomatic deduction, stepwise calculation, and methodological doubt. Hegelian prompting introduces dialectical reasoning through tension, contradiction, and resolution for developing hypotheses through internal contradictions. Humean prompting prioritizes empirical patterns, analogy, and cautious evidence-based generalization. Platonic prompting encourages abstraction and idealized reasoning oriented towards conceptional patterns and abstract reformulation. Kantian prompting emphasizes premises-focused reasoning, formal constraints, and consistency with governing principles [38, 39]. A concise summary of each PA is provided in table 1, with additional methodological detail in Text S5 (operational definition and scope of the PAs) and the full system prompts in Text S6 (system prompts for Each PA).

We note here that these PAs should be understood as philosophy-inspired reasoning templates, not as comprehensive or historically exhaustive reconstructions of the philosophers themselves. Our goal is not to adjudicate philosophical interpretation. Rather, we test whether selected reasoning motifs can be translated into operational system prompts for chemistry question answering. We therefore use labels such as Socratic, Kantian, or Humean as shorthand for specific prompt behaviors, such as assumption testing, rule-based judgment, empirical grounding, abstraction, or structured decomposition. The full prompts are provided in the Supporting Information to make these operational definitions transparent.

The set of Philosophy Agents used here is also not intended to cover the full philosophy-of-science canon. We selected frameworks that could be translated into stable single-agent prompting strategies for direct benchmark answering. Other traditions, including Baconian induction, Popperian falsification, Peircean pragmatism, logical positivist verification, and Kuhnian or Lakatosian accounts of scientific change, are highly relevant but often operate at the level of workflows, hypothesis testing, theory revision, or research-program evaluation. These traditions are therefore better suited for future multi-agent systems with memory, feedback, and external validation.

The core principles used by each agent (‘reasoning tool’) were extracted from their respective system prompts and are summarized in table 1. All agents operated under identical constraints and interacted with the base models in a single-turn setting. This design isolates the effect of philosophical structure on reasoning behavior and avoids confounding factors arising from multi-step orchestration or tool use. We note here that we use the term ‘Philosophy Agent’ to refer to a prompt-conditioned language agent: a base LLM endowed with a persistent system-level reasoning policy that constrains how it approaches a task [49–51]. Because ‘agent’ is used broadly in the recent LLM literature, we distinguish our prompt-conditioned agents from autonomous [52, 53], reasoning-and-acting [54, 55], and tool-using LLM agents [26, 56]. The PAs presented here are not fully autonomous agents, as they do not use external tools, retrieval, persistent memory, multi-turn deliberation, planning, or environment interaction.

2.2. Base LLMs

All promptings were conducted using the Argo API, Argonne National Laboratory's internal interface for LLMs. We evaluated Philosophy Agents on three base architectures: GPT-4o (latest), GPT-5, and GPT-5.1 [47]. These models were selected as a representative set within a single model family to establish a consistent baseline for comparison. While models such as Claude and Gemini may offer complementary capabilities, we focused on GPT models in this initial study to reduce variability and isolate the effects of philosophical prompting. The GPT-4o (snapshot 2024-11-20; referring to the specific model checkpoint version released by OpenAI) represents a strong general-purpose model without explicit long-form reasoning optimization. Its training data extend up to October 2023. The GPT-5 and GPT-5.1 represent two of OpenAI's latest models (while drafting this work) with newer architectures and improved internal reasoning and planning capabilities. GPT-5 was trained on data up to 24 October 2024, while GPT-5.1 was trained up to 30 September 2024. This selection allows us to systematically assess how philosophy-guided prompting interacts with base model strength, ranging from weaker or less stable reasoning behavior to highly capable architectures. We did not evaluate o-series models, as their native multi-step reasoning and internal tool orchestration introduce additional sources of structure that make it difficult to isolate the effect of externally imposed philosophical prompting [47]. We also excluded GPT-5.2, as its training data extends beyond the ChemBench [48] release (Nov 2024), raising concerns about potential data leakage.

2.3. Model execution and inference setup

For each query, a PA system prompt was prepended to the user prompt corresponding to a ChemBench question. Each PA system prompt is a structured document that encodes a philosopher's reasoning framework as a set of principles, reformulation rules, and response guidelines. For example, the Humean system prompt instructs the model to ground its analysis in empirical patterns, treat causal claims as hypotheses built from observation, and weigh evidence in terms of probability and degrees of belief. (table 1). The model responses were collected verbatim and parsed to extract the final numerical answers. No intermediate reasoning steps were removed or edited. To ensure comparability across our agents and models, all runs used identical decoding parameters. The maximum token budget was set to 2000 tokens. The temperature was fixed at 0.2, and the nucleus sampling parameter was set to $\text{top}_p = 0.3$. These values were chosen to limit stochastic variation while allowing sufficient generative flexibility for structured reasoning. Each question (*vide infra*) was evaluated independently for every combination of PA and base model. No adaptive prompting or feedback was applied between questions. This ensures that all reported results reflect intrinsic model-agent interactions rather than cross-question learning effects. Full system prompts and the ChemBench questions used in this study can be found on the project's GitHub repository: https://github.com/HassanHarb92/Sci_reasoning_LLMs.

2.4. Accuracy-error evaluation protocol

To assess model performances, we used the open-ended numerical-answer subset of ChemBench [48]. ChemBench is a benchmark for evaluating the chemical knowledge, reasoning, and calculation abilities of LLMs across a broad range of chemistry topics and skills. It comprises over 2700 curated question-answer pairs spanning undergraduate- and graduate-level chemistry, with expert-validated annotations for task type (knowledge, reasoning, calculation) and chemistry subdomain [48]. While the majority of ChemBench consists of multiple-choice questions, a smaller subset contains open-ended numerical questions that require explicit quantitative reasoning and yield continuous-valued answers. These cover the following chemistry areas: physical chemistry, analytical chemistry, general chemistry, inorganic chemistry, and organic chemistry. ChemBench questions are also annotated by the skills required to answer them. Each question is labeled as knowledge, reasoning, calculation, or a combination of these skills. In this work, we focus exclusively on this numerical subset (243 questions), as it enables a more fine-grained evaluation of model behavior than binary correctness alone. Specifically, numerical answers allow us to quantify how close a model's prediction is to the ground-truth value, rather than only whether it selects the correct option. This distinction is critical for scientific applications, where near-correct predictions may still be informative or actionable. Restricting the analysis to numerical questions also avoids confounding effects introduced by multiple-choice elimination strategies and better isolates the impact of philosophical prompting on structured quantitative reasoning. Representative questions and model responses can be found in the supporting information (Text S2).

Model performance was evaluated using a graded error-tolerance framework rather than strict exact-match accuracy alone. For each question, the model's numerical prediction was compared to the ground-truth reference value provided by ChemBench. (Text S1) The relative error was computed as the absolute deviation normalized by the reference value. We then computed accuracy as the fraction of predictions

Table 2. Statistics of the annotations about the classifications of the task types and chemistry domains of the ChemBench[48] dataset provided by its developers.

ChemBench annotation	Question count	Percentage
Task Type		
reasoning	183	67.8%
calculation	65	24.1%
knowledge	23	8.5%
Subdomain		
physical chemistry	67	24.7%
analytical chemistry	50	18.5%
general chemistry	47	17.3%
Inorganic chemistry	37	13.7%
organic chemistry	33	12.2%

falling within predefined relative error thresholds of 1%, 5%, 10%, 15%, and 20%. This Accuracy-vs-Error analysis captures not only whether a model produces correct answers, but also how close incorrect answers are to the true value. Such distinctions are particularly important for scientific applications, where near-correct numerical predictions may still be actionable. All accuracy values were computed independently at each threshold. Model responses were aggregated across task types and chemistry subdomains using the annotations provided by the ChemBench developers. Task types refer to reasoning, calculation, and knowledge. Chemistry subdomains refer to physical, analytical, general, inorganic, and organic chemistry (table 2). The task types and chemistry domains of the dataset is based on the annotation of the same dataset with its developers [48].

2.5. Answer extraction and evaluation

To ensure consistent and unbiased extraction of numerical answers from model outputs, we employed a single proxy LLM agent for post-processing across all experiments. This proxy agent was used exclusively to identify and extract the final numerical prediction from each generated response and did not participate in reasoning or problem solving. All extractions were performed using GPT-5.1 with a temperature of 0.0 to guarantee deterministic behavior. The same extraction prompt and parsing logic were applied uniformly across all PAs, base models, task types, and chemistry subdomains. This design choice ensures that any observed performance differences arise from the reasoning behavior induced by the philosophical system prompts rather than from variability in answer parsing or interpretation. The use of a fixed, deterministic proxy model isolates evaluation from generation and provides a controlled mechanism for handling heterogeneous response formats produced by different philosophical strategies. Because GPT-5.1 is also one of the evaluated generation models, we acknowledge a possible extractor-model coupling. However, the proxy model was used only for deterministic post-processing of final numerical values and did not participate in reasoning, answer generation, or task solving. The same extraction protocol was applied uniformly across all base models and PAs.

3. Results

We first looked at the overall performance of the agents and compared them to base LLM models. Figure 1 compares the performance of Philosophy Agent-based models against their corresponding base models across three architectures: GPT-4o, GPT-5, and GPT-5.1. The top row reports accuracy within a strict 1% relative error threshold. At this level, the model's prediction deviates by no more than 1% from the reference value. This is effectively quantitative reliability for most chemistry applications. The bottom row of figure 1 shows accuracy as a function of increasing error tolerance from 1% to 20%. Together, these views provide both a high-precision snapshot and a broader assessment of near-correct prediction behavior.

Across all three base models, PAs generally outperformed their corresponding base models (see below). This trend is already evident at the 1% threshold, indicating that PA reasoning improves precision rather than merely increasing tolerance to error. However, the magnitude and consistency of these improvements depend strongly on the underlying base model and the specific agent.

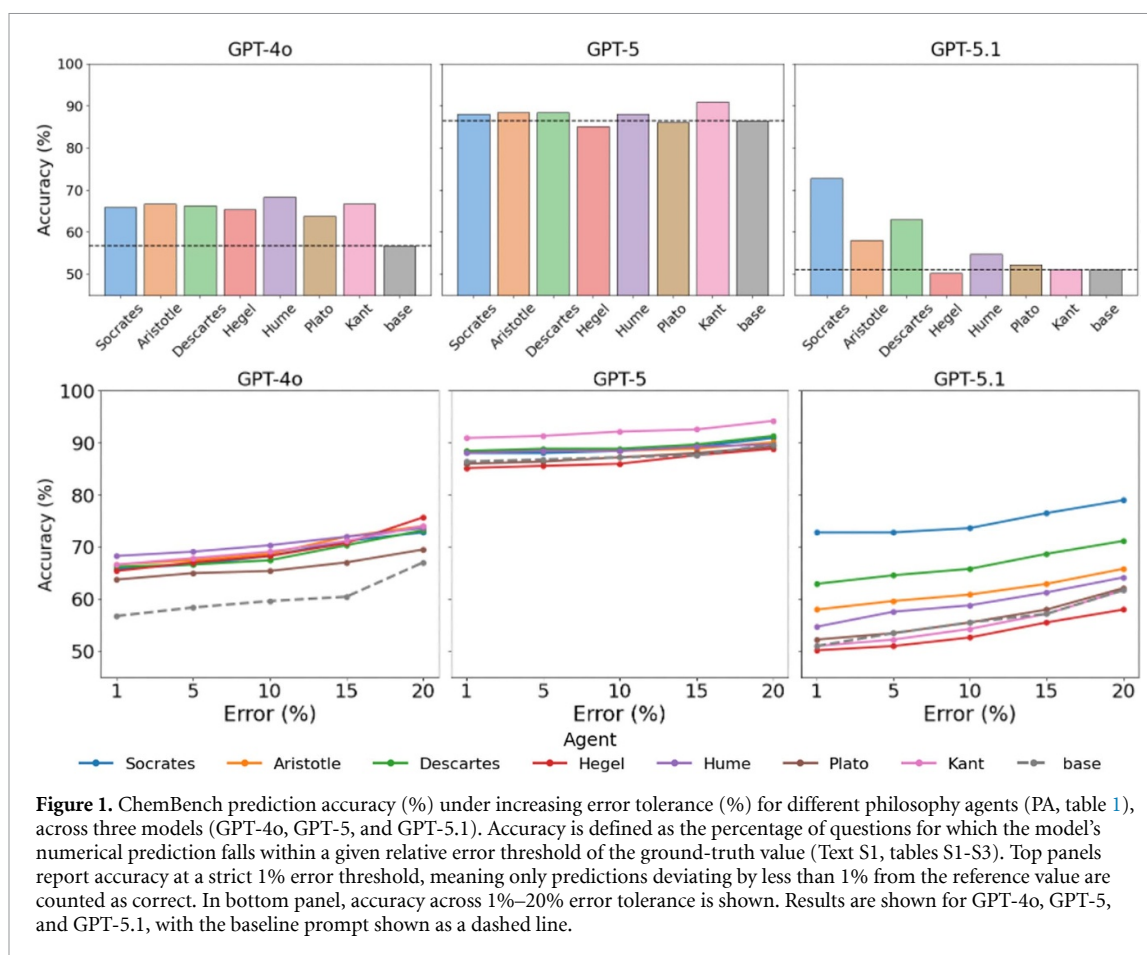
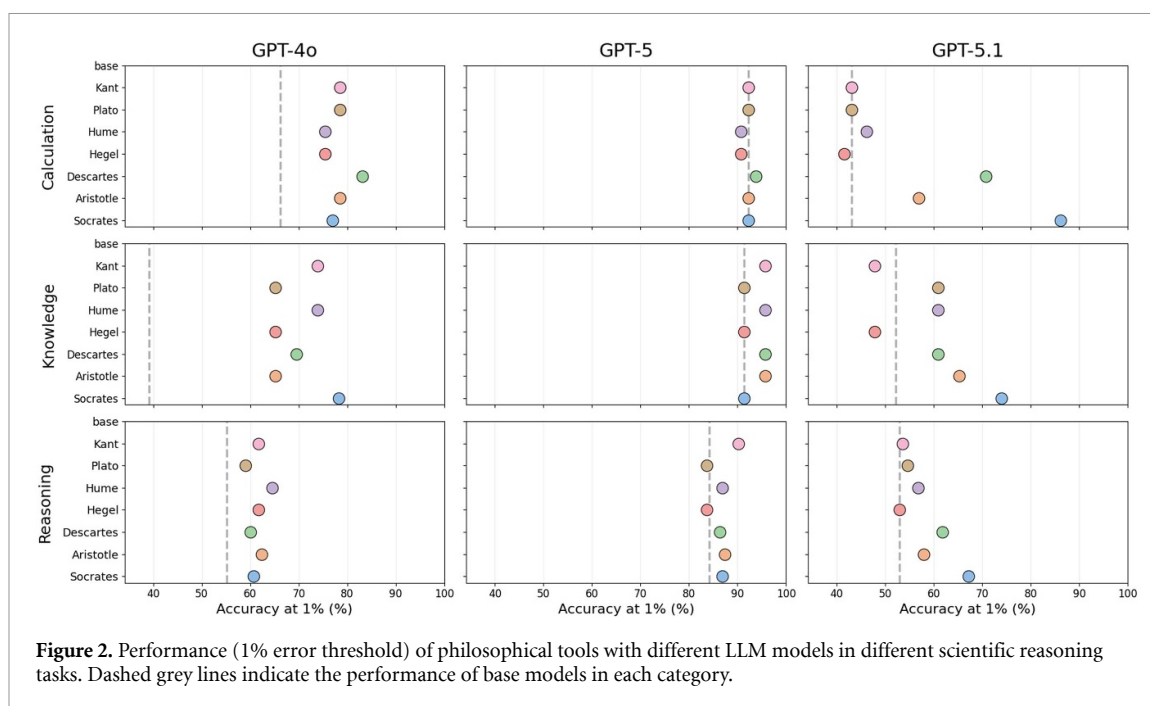


Figure 1. ChemBench prediction accuracy (%) under increasing error tolerance (%) for different philosophy agents (PA, table 1), across three models (GPT-4o, GPT-5, and GPT-5.1). Accuracy is defined as the percentage of questions for which the model's numerical prediction falls within a given relative error threshold of the ground-truth value (Text S1, tables S1-S3). Top panels report accuracy at a strict 1% error threshold, meaning only predictions deviating by less than 1% from the reference value are counted as correct. In bottom panel, accuracy across 1%–20% error tolerance is shown. Results are shown for GPT-4o, GPT-5, and GPT-5.1, with the baseline prompt shown as a dashed line.

When using GPT-4o model (figures 1(a) and (d)), all PAs outperformed the base model at the 1% threshold. Agent accuracies exceeded the baseline by a range of 7–11 percentage points. Among PAs, Hume exhibited the highest accuracy (68.3%), followed by Aristotle and Kant (both 66.7%). The corresponding threshold curves in figure 1(d) show that these gains persisted across the full range of error tolerances. The curves remained approximately parallel to the baseline, indicating that PAs primarily produced an upward shift in accuracy rather than altering the sensitivity to error threshold. This suggests a systematic improvement in prediction quality across difficulty levels.

On the other hand, when GPT-5 model was used (figures 1(b) and (e)), absolute performance was substantially higher, reflecting the strength of the base model. The base model achieved an accuracy of 86.4% at the 1% threshold, compared to 56.8% for GPT-4o and 51.0% for GPT-5.1. In this higher-accuracy regime, agent-specific differences were more compressed. Most PAs continued to outperform the base model, with Kant achieving the highest accuracy (90.95%, 11 more questions answered correctly), followed by Aristotle and Descartes (both approximately 88.5%). Two exceptions are observed at the strict 1% threshold: Hegel (85.2%) and Plato (86.0%) slightly underperformed relative to the base model. Importantly, the threshold curves in figure 1(e) show that these underperformances were limited in magnitude and did not dominate behavior at higher error tolerances, where most agents converged toward or exceeded the base model.

Furthermore, when GPT-5.1 model was employed (figures 1(c) and (f)), PA reasoning has the largest relative impact. Notably, GPT-5.1 underperformed GPT-4o and GPT-5 despite being a newer release. The exact cause is unclear, as model architectures and training details are not publicly disclosed. However, this likely reflects internal tradeoffs in instruction tuning that reduced reliability on multi-step, domain-specific reasoning tasks. The base model defined a clear lower bound across all thresholds, with an accuracy of 51.0% at the 1% threshold. In contrast, several PAs achieved substantial improvements. Socrates exhibited the strongest performance, reaching 73.1% accuracy at the 1% threshold, representing an improvement of over 22 percentage points relative to the base model. Descartes (63.0%) and Aristotle (58.0%) also showed marked gains. Hegel again underperformed relative to other agents and remained close to the baseline. The divergence between agent-guided and base curves in figure 1(f) increases with error tolerance, indicating that explicit reasoning structure was particularly effective at mitigating systematic error accumulation in weaker base models.



1. Based on the results shown in figure 1 reveals three consistent patterns in how PAs influence model performance. First, philosophy-guided reasoning consistently enhances numerical accuracy across a wide range of error thresholds, demonstrating its general effectiveness. Second, the degree of improvements is tied to the capability of the base model—with lower-performing models benefiting the most. Third, effectiveness of agent is not universal; the relative performance of each agent varies by model, highlighting the importance of alignment between reasoning strategy and model architecture. For example, at the strict 1% error threshold, Socratic prompting yields a +21.8 percentage-point gain with GPT-5.1, Kantian prompting yields a +4.5 percentage-point gain with GPT-5, and Humean prompting yields a +11.5 percentage-point gain with GPT-4o.

3.1. Performance by Task Type

To evaluate how philosophical prompting interacts with different types of ChemBench tasks, we grouped the questions into three task types based on annotations provided by Jablonka and coworkers. From their 243 numerical questions, 183 were labeled as reasoning, 65 as calculation, and 23 as knowledge-based (table 2). Figure 2 presents accuracy across these categories at a strict 1% error threshold, with dashed lines marking the base model scores. Table 3 highlights the top-performing agent in each case. A detailed breakdown of questions with overlapping task annotations and all model scores is provided in the Supporting Information (tables S1 and S2).

Calculation: Philosophical prompting consistently improved accuracy on calculation questions, with the most dramatic effects seen in GPT-5.1. The Socratic agent nearly doubled its performance (43.1%→86.2%), followed by Descartes and Aristotle. GPT-4o also benefited from all agents, with Descartes achieving the highest score at 83.1%. For GPT-5, baseline performance is good (84%), and Descartes again shows accuracy (93.8%).

Knowledge: Knowledge-based tasks showed strong gains for GPT-4o and GPT-5.1, particularly under Socratic prompting, which raised GPT-4o's accuracy from 39.1% to 78.3%. Hume and Kant also performed well, highlighting the value of grounding and structured logic. GPT-5 reached 95.7% with several agents, including Aristotle, Descartes, and Hume. For GPT-5.1, Socrates again performed best (73.9%), while other agents had mixed outcomes.

Reasoning: Reasoning tasks produced moderate but consistent improvements across models. For GPT-4o, Hume led with 64.5%, followed by Aristotle and Kant. GPT-5 reached its peak with Kantian prompting (90.2%). GPT-5.1 responded best to Socratic prompting (67.2%, a 14-point gain), while other agents yielded smaller gains.

Table 3. Top performing Philosophy Agents (PA) for each task type and model.

Task type	Model	Top PA	Top PA accuracy (%)	Base model accuracy (%)
Calculation	GPT-4o	Descartes	83.1	66.2
	GPT-5	Descartes	93.8	92.3
	GPT-5.1	Socrates	86.2	43.1
Knowledge	GPT-4o	Socrates	78.3	39.1
	GPT-5	Aristotle/Descartes/ Hume/Kant	95.7	91.3
	GPT-5.1	Socrates	73.9	52.2
Reasoning	GPT-4o	Hume	64.5	55.2
	GPT-5	Kant	90.2	84.1
	GPT-5.1	Socrates	67.2	53.0

Across all task types, PAs generally outperformed their corresponding base models, with the magnitude of improvement depending strongly on both the task and the underlying architecture. Gains were largest for weaker base models, particularly GPT-5.1, and most pronounced in calculation and knowledge tasks. Among the agents tested, Socrates (basic critical thinking), Descartes (axiomatic deduction), Kant (premises-prioritized reasoning), and Hume (empirical grounding) emerged as consistent top performers, though no single strategy dominated universally. We note here that GPT-5.1 underperformed relative to GPT-5 despite being a newer model. The exact cause is unclear due to undisclosed model architectures. However, this likely reflects tradeoffs in architecture or instruction tuning that reduced reliability on multi-step, domain-specific reasoning tasks. The weaker performance of GPT-5.1 on knowledge tasks further suggests less stable internal factual recall. At the same time, GPT-5.1 exhibited greater sensitivity to philosophical prompting, suggesting that external structure can help compensate for weaker internal planning behavior.

3.2. Performance by chemistry subdomain

Next, we examine model and agent performance across five chemistry subdomains: physical, analytical, general, inorganic, and organic chemistry. These subdomains were also retrieved from ChemBench's expert annotations shown in table 2. Accuracy results are shown in figure 3 with top-performing agents summarized in table 4. Full breakdowns are available in supporting information (table S3).

Physical Chemistry: Philosophical prompting yielded robust gains in physical chemistry, particularly for GPT-5.1. Socrates improved accuracy from 56.7% to 94.0%, while Descartes and Aristotle also showed strong effects. GPT-4o benefitted across all agents, with Descartes reaching 86.6%. GPT-5's baseline (97.0%) left little room for improvement, though Descartes slightly outperformed it. Agent effectiveness was especially pronounced for models with lower baseline stability.

Analytical Chemistry: This subdomain posed consistent challenges. GPT-4o performed poorly (16.0%) and showed no improvement across agents. GPT-5 reached 72.0% baseline, with Kant, Socrates, and Hume providing small gains. GPT-5.1 had the lowest baseline (8.0%) but responded to Socrates and Plato with 20.0%—a 12-point improvement—indicating that even in difficult domains, structured prompting can partially recover reasoning ability.

General Chemistry: All models benefitted from prompting in general chemistry. GPT-4o saw the largest improvements, with Hume boosting accuracy from 63.8% to 93.6%. GPT-5 showed selective gains, with Kant leading at 91.5%. GPT-5.1 again responded best to Socrates (89.4%) and Aristotle (87.2%), reinforcing their utility across low-stability settings.

Inorganic Chemistry: This subdomain had high baseline accuracy for GPT-4o (89.2%) and GPT-5 (97.3%). For both, prompting offered small but consistent gains, led by Aristotle and Descartes. GPT-5.1 showed more variation, with Socrates reaching 91.9% (up from 70.3%) and Descartes following at 86.5%.

Organic Chemistry: Organic chemistry revealed high model dependence. GPT-5 reached near-perfect scores with several agents, including Aristotle and Descartes (100.0%). GPT-4o showed small, uneven

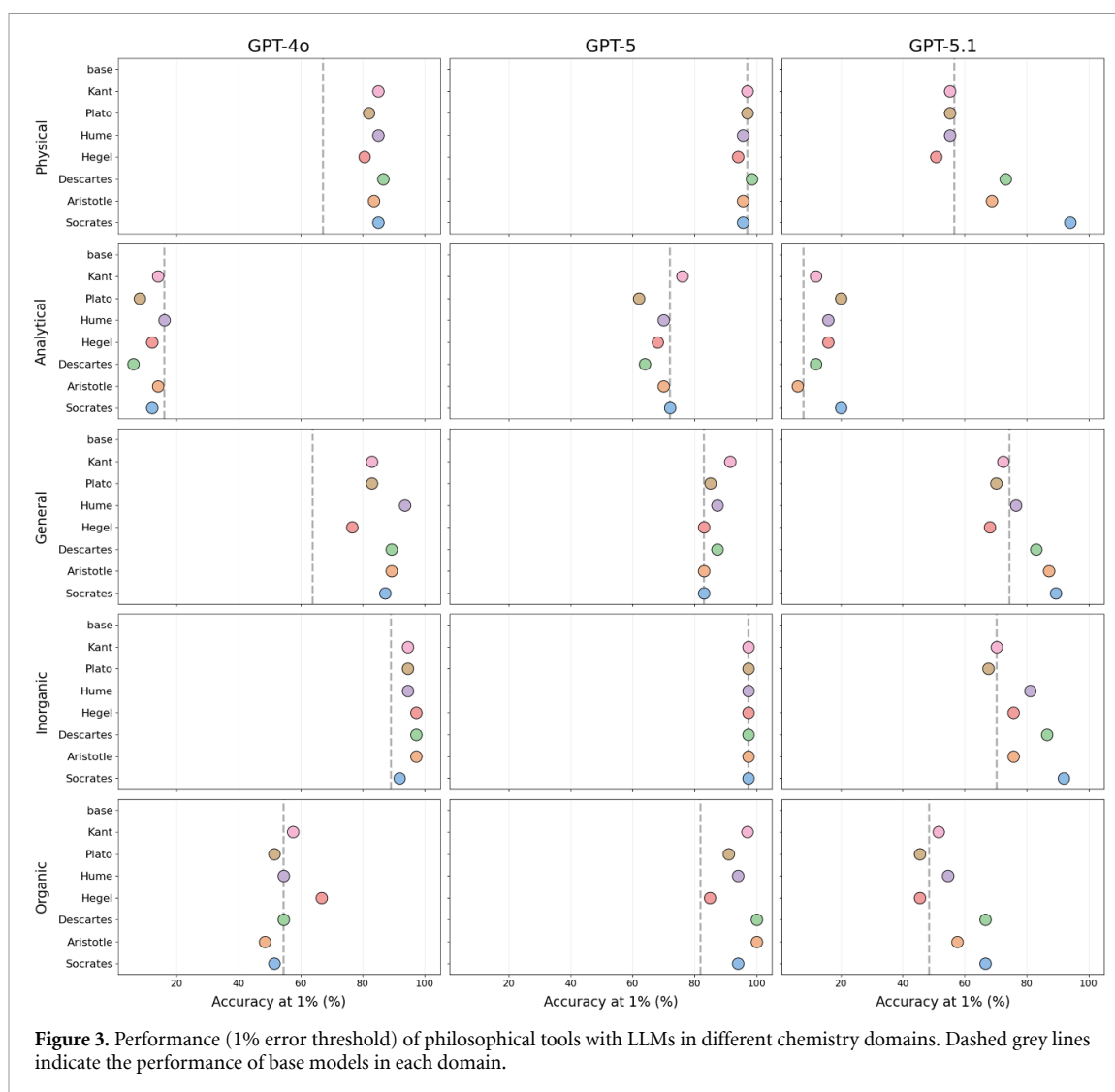


Table 4. Top performing agent by chemistry subdomain and model.

Subdomain	Model	Top PA	Top PA accuracy (%)	Base Model accuracy (%)
Physical	GPT-4o	Descartes	86.6	67.2
	GPT-5	Descartes	98.5	97.0
	GPT-5.1	Socrates	94.0	56.7
Analytical	GPT-4o	—	16.0 (base)	16.0
	GPT-5	Kant	76.0	72.0
	GPT-5.1	Socrates	20.0	8.0
General	GPT-4o	Hume	93.6	63.8
	GPT-5	Kant	91.5	83.0
	GPT-5.1	Socrates	89.4	74.5
Inorganic	GPT-4o	Aristotle/Descartes/Hegel	97.3	89.2
	GPT-5	All agents	97.3 (tie)	97.3
	GPT-5.1	Socrates	91.9	70.3
Organic	GPT-4o	Hegel	66.7	54.5
	GPT-5	Aristotle/Descartes	100.0	81.8
	GPT-5.1	Socrates/Descartes	66.7	48.5

gains; Hegel performed best at 66.7%. GPT-5.1 again showed strong sensitivity to prompting, with Socrates and Descartes improving from 48.5% to 66.7%.

Within chemistry subdomains, PAs generally improved model performance, with the extent of gains varying by both domain complexity and model stability. The largest improvements again appeared

for GPT-5.1, up to +38.5% in general, +36.3% in physical, and +27.1% in inorganic chemistry. In these cases, Socrates and Descartes PAs were among the most effective PAs. GPT-5 showed strong baselines across all subdomains, leaving less room for improvement, while GPT-4o demonstrated moderate but consistent gains. We note that performance varied substantially across subdomains, with analytical and organic chemistry posing greater challenges across all models. Specifically, PAs showed problem-dependent behavior rather than a single universal pattern. Physical, general, inorganic, and organic chemistry showed clear gains for at least one Philosophy Agent, while analytical chemistry remained difficult across models. These trends highlight that the impact of philosophical prompting depends not only on the model architecture but also on how well the agent's reasoning style aligns with the structure of the underlying scientific domain. We note that several subdomain and task-type groups are relatively small, especially organic chemistry, inorganic chemistry, and knowledge questions. As a result, small differences in tables 3 and 4 should be interpreted cautiously, since a single question can shift the reported accuracy by several percentage points. Analytical chemistry was also consistently difficult for GPT-4o, with low accuracy across the base model and all Philosophy Agents. This suggests that some subdomains may require information or structure that is not recovered by prompting alone. This limitation could be explored in future work using larger benchmark subsets.

4. Discussion

The task- and domain-resolved analysis revealed that philosophical prompting is not a uniform enhancement mechanism, but rather a conditional scaffold whose effectiveness depends on both the underlying model architecture and the cognitive demands of the task. In general, for direct pattern-fulfillment models (e.g. GPT-4o), distinct philosophical strategies emerge as optimal for different cognitive regimes, indicating that no single reasoning style is universally dominant. For common models with internal response refinement architecture, iterative questioning and explicit epistemic grounding can act as a stabilizing mechanism when the model lacks robust internal structure. In contrast, newer and bigger models (such as GPT-5) exhibit smaller and more selective improvements, with gains concentrated in reasoning tasks under Kantian-style premises-prioritized structuring and in calculation tasks under Cartesian decomposition.

The general performances reflect the 'personality' of different base models (figure 1). When handling near-miss predictions, GPT-4o had a significant jump of accuracy between 15% and 20% error, indicating that this very fast model already on the right track for most of the chemistry cases. GPT-4o, by nature (less reliably self-correcting without external scaffolds), showed lacking all philosophical reasoning capabilities. The significant improvement came from Hegel for most of the edge case with Hume improved the 'near-miss' cases. This reflects that GPT-4o was just guessing the answers based on flat context without proper evidence gathering (Hume) or contradiction reflection (Hegel). This behavior is further reflected in figure 2, where the most significant improvement happening in the 'Knowledge' category of all the philosophical tools with GPT-4o. Furthermore, GPT-5 base model (figure 1) performed generally well for ChemBench tasks, with the most improvement came from Kant focusing on the systematic construction of premises of arguments. This behavior was further justified in the performance of philosophical tools in different categories (figure 2), where Kant model showed significant improvement in the 'Reasoning' category of GPT-5. In contrast, GPT-5.1 exhibited weaker base model performance than GPT-4o across all task categories (figure 1), suggesting lower default reliability. Yet, it consistently showed the largest gains when paired with PAs (figure 2), especially in the 'Calculation' category. This implies that while GPT-5.1 may struggle with internal planning or multi-step reasoning, it is more responsive to structured external prompting. PAs like Socrates and Descartes were particularly effective, reflecting GPT-5.1's ability to follow guided critical thinking and deductive strategies, even when it lacks robust reasoning out-of-the-box.

The performances of all PAs justify that scientific research, by-nature, is a multi-agent approach requires different philosophical tools at different stages. The difference in domain-specific accuracy across different scientific domains and philosophical tools indicates that the capability of thinking chemistry problems at a more abstract level still needs to be improved despite the scoring on fulfilling patterns of the chemistry problems. Furthermore, those philosophical tools, as abstraction of reasoning patterns beyond direct pattern fulfillment, might be interesting criteria to evaluate and develop reasoning capability of LLMs. Perhaps in addition to training set and scores, a reasoning pattern from those philosophy schools needs to be enforced via system prompts to observe the capability of a reasoning model following different reasoning patterns during training. A qualitative case study is provided in the Supporting Information (Text S3), where we analyze an analytical chemistry NMR question to test

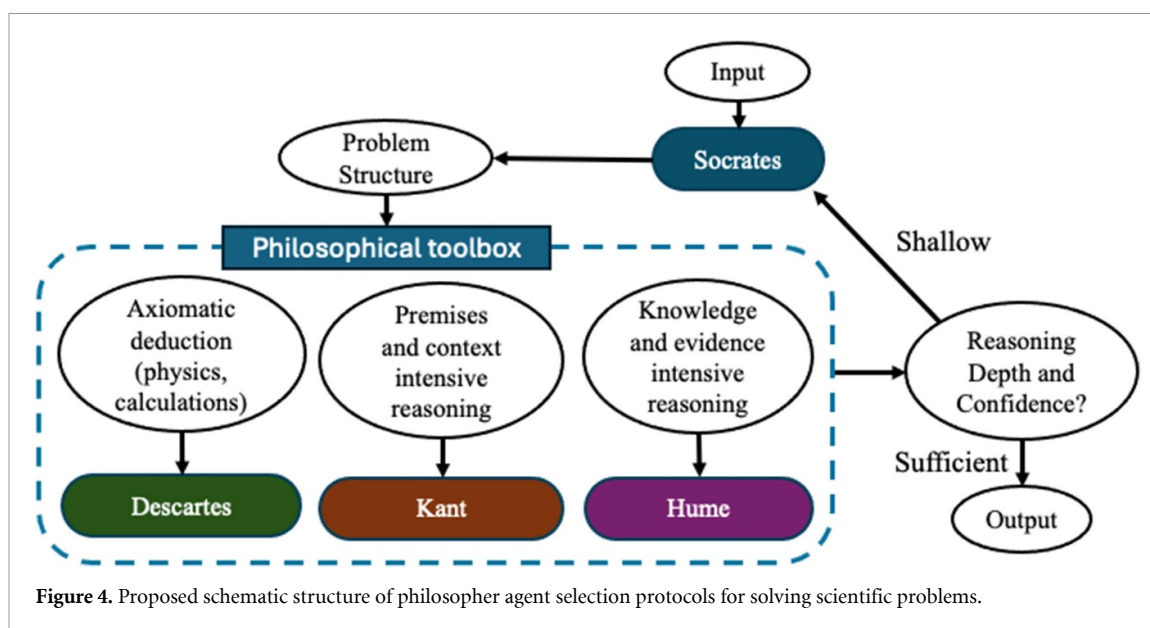


Figure 4. Proposed schematic structure of philosopher agent selection protocols for solving scientific problems.

whether Philosophy Agents follow their intended reasoning principles and how correct and incorrect reasoning pathways differ. Additionally, to test whether the observed gains reflect philosophy-inspired prompting specifically or structured prompting more generally, we added two non-philosophical controls: a generic reasoning prompt and a chemistry-expert prompt. These results are reported in the Supporting Information (Text S4 and table S4).

4.1. Three rules of thumb

Based on our results, we propose a set of three rules of thumb to serve as guidelines in choosing a proper philosopher agent. These guidelines are summarized schematically in figure 4, which provides a practitioner-facing decision flow for selecting philosophical agents.

4.1.1. Align philosophical guidance with base model capability

For smaller base models (i.e. GPT-4o) that need boost on basic critical thinking capability, structured philosophical prompting (particularly Socratic-style guidance) provides substantial gains by enforcing explicit reasoning steps and reducing error propagation. For stronger base models, overall performance is already high, and PAs offer smaller but more targeted improvements that depend on task structure rather than general reasoning strength. Regardless, philosophical tools for the LLM reasoning needs to be tailored based on the capability and tendencies of the each base model, but those philosophical tools must exist in some form to achieve structured reasoning that required for insightful scientific reasoning beyond average pattern fulfillment.

4.1.2. Match philosophical strategy to the dominant cognitive demand

Different PAs excel at different forms of reasoning. For example, Descartes-style prompting is most effective for calculation-heavy and physically grounded problems, such as Hess's law enthalpy calculations that require decomposing multiple thermochemical equations and combining them stepwise. On a B_2H_6 combustion problem, Descartes was the only agent to reach the correct enthalpy of 1152.4 kJ, while the base GPT-5 model and all other agents failed. Kant-style prompting benefits tasks requiring rule-based reasoning and structured logical constraints. An example is the total electron counting in molecules like SF_6 , where the answer follows directly from summing atomic numbers. Here, Kant uniquely recovered the correct count of 70 electrons while the base GPT-5 model returned 48. Hume-style prompting performs best for empirically driven or example-oriented questions where inductive generalization plays a central role. This is illustrated by predicting 1H NMR peak counts, which depends on recognizing symmetry patterns from prior examples. For a cycloheptadiene carbaldehyde, Hume was the only agent to correctly predict 7 peaks, while the base model reported 4. Socratic prompting, grounded in iterative questioning and self-examination, was particularly effective at rescuing weaker base models from catastrophic errors. On GPT-5.1, Socrates correctly identified the oxidation state of hydrogen (+1) in the products of the Sabatier reaction, while the base model returned -2. Similarly, Aristotelian prompting, with its emphasis on first-principles categorization, uniquely solved a $CaSiO_3$ acid-balancing problem ($x = 10$) that stumped GPT-4o and every other agent. Ideally, those philosophical tools (instead of

apparent text patterns of knowledge) need to be dynamically applied for composite and complicated scientific problems.

4.1.3. Avoid universal prompting strategies

No single PA consistently outperforms others across all base models, task types, and chemistry subdomains. Optimal performance requires conditional selection of prompting strategies based on both the underlying model and the nature of the task, rather than the application of a single, fixed philosophy across all scenarios.

We emphasize that figure 4 is intended as a conceptual roadmap rather than a validated multi-agent workflow. The present study evaluates single-agent Philosophy Agent systems, where each reasoning style is tested independently under controlled conditions. Testing dynamic agent selection, agent routing, and multi-agent deliberation will require a separate experimental framework. Such systems represent an important direction for future work, where the single-agent reasoning signatures identified here can be used to guide more adaptive multi-agent LLM reasoning workflows.

Several questions emerge from our findings that require further investigation to improve scientific reasoning with LLMs. The consistently larger gains observed for smaller base models suggest that philosophy agents act, at least in part, as external stabilizers that impose structure where the underlying model struggles to maintain coherent reasoning. Understanding which aspects of Socratic prompting confer this robustness, such as explicit step decomposition, self-questioning, or hypothesis elimination, remains an important area for deeper analysis. Broadly, the observed specialization of Descartes-, Kant, and Hume-style agents across different reasoning regimes raises the question of whether these effects are primarily driven by prompt structure or by how such structures interact with latent reasoning capabilities already present in the base models. Furthermore, deciding the criteria for which philosophical tools should be used for what task, and how to test/train the model to use them, becomes a complicated question beyond the simple data extraction, classification, and test dataset design.

Future work should also examine the generality and limitations of philosophy-guided prompting beyond ChemBench. Promising directions include testing on broader chemistry benchmarks such as MatBench [57, 58] for materials property prediction, SciQA [59] for open-ended scientific question answering, and domain-specific reasoning sets in biology, physics, and medicinal chemistry [60]. While the current study spans multiple task types and chemistry subdomains, it remains to be tested whether similar agent–task alignments hold across other scientific benchmarks, domains, and evaluation criteria. We also note that the numerical subset of ChemBench used here comprises only 243 questions, which limits statistical power and coverage. Larger and more diverse datasets, such as full multi-thousand-question benchmarks spanning reaction prediction, spectroscopy interpretation, and reaction mechanism elucidation, will be required to assess robustness more rigorously. Additionally, the existence of clear failure modes when certain philosophical strategies are applied (e.g. Hegel, Plato) highlights the need for adaptive or automated agent selection. By ‘adaptive agent selection,’ we mean a system that automatically chooses the best philosophical prompting strategy for each question, rather than applying a single fixed agent to all tasks. We did not implement this in the current study because our goal was first to characterize how each agent performs in isolation, establishing the baseline behavior needed to inform future selection strategies. Rather than relying on fixed heuristics (i.e. predetermined rules such as ‘always use Descartes for calculation questions’), future systems could analyze an incoming question, identify whether it requires calculation, rule-following, pattern recognition, or critical questioning, and then route it to the most suitable PA.

Further work can also include combining multiple agents when the question demands mixed reasoning styles. Future work will also include prompt-mechanism ablations that directly test whether performance depends on philosopher names, coherent philosophical principles, or structured reasoning more broadly. These studies will include name-permuted prompts, where the reasoning principles are kept fixed but the philosopher label is changed; incoherent philosophical prompts, where philosophical vocabulary is retained but the reasoning structure is disrupted; and stacked-philosopher prompts, where two or more reasoning styles are combined. These tests are distinct from the present single-agent benchmark and are needed to separate the effects of philosophical identity, prompt structure, and reasoning-principle alignment.

5. Conclusion

In this work, we systematically evaluated philosophy-guided agentic prompting as a mechanism for improving numerical reasoning in chemistry-focused tasks using LLMs. We benchmarked seven philosophy agents (PA) across multiple base architectures, task types, and chemistry subdomains using a quantitative error-based metric. Philosophical prompting delivered substantial improvements in accuracy, particularly under strict error thresholds. Gains at the strict 1% error threshold include +21.8 percentage points for GPT-5.1 with Socrates, +11.5 percentage points for GPT-4o with Hume, and +4.5 percentage points for GPT-5 with Kant, relative to their corresponding base models. However, these gains are neither uniform nor universal. Instead, the effectiveness of a given philosophical strategy depends strongly on the underlying model capability and the dominant cognitive demands of the task.

Our results demonstrate that philosophy-inspired agents function best as conditional scaffolds rather than as general-purpose enhancements. For smaller models, structured prompting, especially Socratic-style reasoning, provides large and consistent improvements by stabilizing inference and enforcing explicit reasoning steps. For stronger models, gains are more selective and task-dependent, with Cartesian, Kantian, and Humean strategies excelling in calculation-heavy, rule-based, and empirically driven settings, respectively. No single PA dominates across all scenarios, underscoring the need for adaptive agent selection rather than universal prompting strategies.

Overall, this study reframes philosophical prompting not as a stylistic choice, but as a design variable in scientific AI systems. By explicitly aligning prompting strategies with model strength and task structure, philosophy-guided agents can serve as modular, interpretable components for improving reasoning performance in chemistry and beyond. Furthermore, the reasoning capability of a model may need to be evaluated during usage or training with different reasoning patterns ('philosophical tools') imposed. We anticipate that these insights will inform the development of next-generation agentic workflows that integrate reasoning structure, domain knowledge, and model capability in a principled and task-aware manner.

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Data availability statement

The data that support the findings of this study are openly available at the following URL/DOI: https://github.com/HassanHarb92/Sci_reasoning_LLMs [61].

Supplementary information available at <https://doi.org/10.1088/2632-2153/ae792d/data1>.

Author contributions

H. Harb, Y. Sun, and M. Unal developed the concepts. N Chia, Z. Yang, B. Ingram, and R. S. Assary supervised the project and offered guidance. All authors reviewed and approved the final version of the manuscript.

Conflict of interest

The authors declare no competing conflicts of interest.

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